The universal property of 0-truncation: maps from the 0-truncation of A to a set B are maps from A to B

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:trunc0-lump' (source lines 1120-1124, Lemma 6.9.2); the proof environment begins on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own lead-in sentence, which is what connects the counted lemma to the induction principle that its proof applies twice.

Editorial note. The retained counter values (chapter 6, section 9, theorem offset 1) make the retained lemma print as Lemma 6.9.2, matching the original book numbering.

Editorial note. The single external reference points at the induction principle of 0-truncation, which the retained proof invokes by label; the counted proof itself is retained in full.

Editorial scope. One original result of the HoTT Book, counted once: the universal property of 0-truncation (Lemma 6.9.2).

Allowed standard prerequisites (editorial).

- Induction principle of 0-truncation. For B : trunc0 A -> type with g : prd a:A B(|a|_0), assuming each B(x) is a set, there is f : prd x:trunc0 A B(x) with f(|a|_0) := g(a).. It is applied twice: once in the constant-family form at source line 1126 to build the map from right to left, and once at source line 1127 to build h'.
- The 0-truncation is a set. trunc0 A is a set, so its path types are mere propositions.. Together with B being a set it makes the goal h(x) = h'(x) at source line 1130 a mere proposition.

Source reference. Original lemma 6.9.1: sources/hott-hits/pinned-original-source.tex, source lines 1108–1108. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

This implies the expected universal property.

Lemma 6.9.2. *For any set B and any type A, composition with $|-|_0 : A \to \|A\|_0$ determines an equivalence*

$$(\|A\|_0 \to B) \;\simeq\; (A \to B).$$

Proof. The special case of original lemma 6.9.1 when B is the constant family gives a map from right to left, which is a right inverse to the "compose with $|-|_0$" function from left to right. To show that it is also a left inverse, let $h : \|A\|_0 \to B$, and define $h' : \|A\|_0 \to B$ by applying original lemma 6.9.1 to the composite $a \mapsto h(|a|_0)$. Thus, $h'(|a|_0) = h(|a|_0)$.

However, since B is a set, for any $x : \|A\|_0$ the type $h(x) = h'(x)$ is a mere proposition, and hence also a set. Therefore, by original lemma 6.9.1, the observation that $h'(|a|_0) = h(|a|_0)$ for any $a : A$ implies $h(x) = h'(x)$ for any $x : \|A\|_0$, and hence $h = h'$. □